PHYSICAL GEOGRAPHY / ASTRONOMY

The Solar System

SOLAR SYSTEM

The Sun • Planets • Small Bodies • The Moon

Component: Astronomy basics for General Science / Geography sections

🔴 Headings & Definitions 🔵 Explanations 🟡 Highlighter = Key Facts 🎯 Exam Tip ❌ Common Mistake 🧠 Memory Trick

StudyPlanner

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The Sun — Our Star

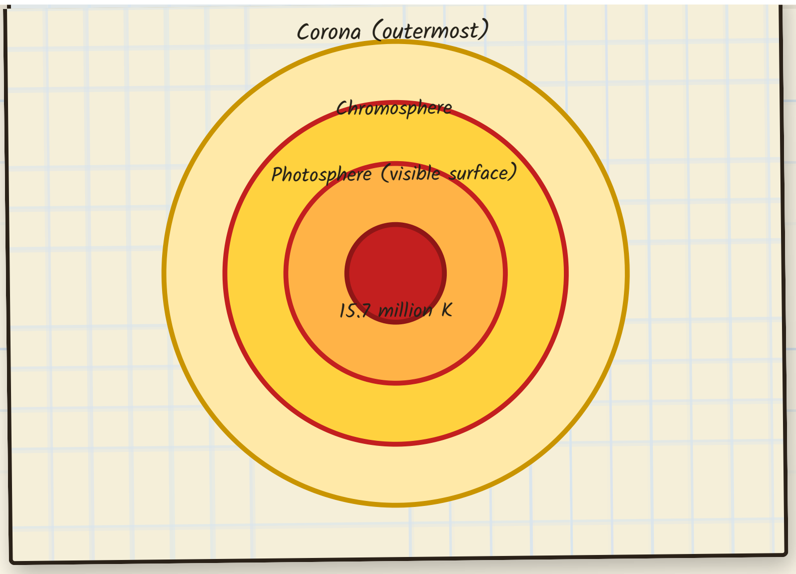
Before the planets, let's understand the **Sun** — a type of **star**. Stars generate their own energy and their own light, so they are called luminous objects. Objects with no light of their own are **non-luminous** — e.g. the **Moon**, which only reflects the Sun's light, which is why it shines at night.

> ❌ Common Mistake
> Don't think the Moon "glows" on its own — it's a **non-luminous** object; it only reflects sunlight.

Layers of the Sun (outside → inside)

- **Corona** — the outermost layer of the Sun
- **Photosphere** — the visible surface of the Sun
- **Chromosphere** — the layer just above the photosphere
- **Core** — the centre of the Sun, temperature ≈ 15.7 million Kelvin

🧠 Fig 1: Layers of the Sun



🧠 *Memory Trick*

Peel the Sun like an onion, centre outward: **Core** → **Chromosphere** → **Photosphere** → **Corona**. (Photosphere = "photo" = the one we can visibly "photograph"; Corona = the outer "crown".)

Measuring Cosmic Distances

- Sunlight takes about 8.2 minutes (i.e. **8 minutes 20 seconds**) to reach Earth's surface
- **Light year** — the distance light travels in one year — 9.46×10^{12} km
- Earth–Sun distance ≈ **150 million km** (precisely, **149.6 million km**)
- **Parsec** — a unit larger than the light year; **1 parsec = 3.26 light years**

❌ *Common Mistake*

Don't confuse **Parsec** (a unit of astronomical distance) with **Pascal** — that's a unit of *pressure*, a completely different thing!

Sunspots

Sunspots are the dark patches we see on the Sun's surface — they appear dark because they mark regions where the temperature becomes extremely high.

📌

🧠 *Quick Recap*

Corona (outer) → Photosphere (visible surface) → Chromosphere (above photosphere) → Core (15.7 million K, centre)!

Terrestrial Planets I — Mercury + Venus

1️⃣ Mercury

- **Closest planet to the Sun**
- **Shortest revolution time** of all planets — one revolution ≈ 88 days
- **No atmosphere** — the Sun's solar winds are strong enough to strip away any atmosphere it might have had
- **No satellites/moons** at all

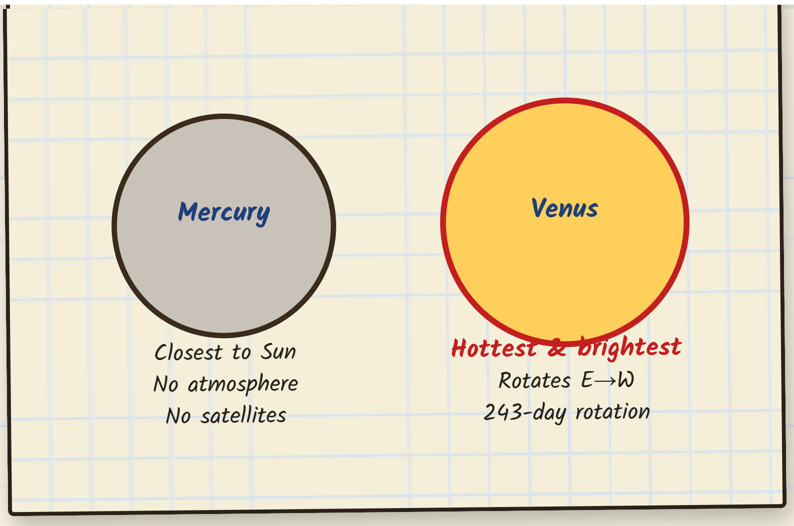
❌ Common Mistake

Don't assume Mercury is the **hottest** planet just because it's closest to the Sun — it is NOT. (Venus is the hottest — see below!)

2️⃣ Venus

- Also called "**Lucifer**" — meaning "Light Bringer"
- Called **Earth's twin planet** — almost the same shape and size as Earth
- Also known as the **Morning Star** and **Evening Star** (Hindi: Bhor ka Tara and Sanjh ka Tara)
- The **hottest planet** AND the **brightest planet**
- Hottest because of abundant CO_2 (a greenhouse gas that traps heat very effectively) and thick clouds of H_2SO_4 (sulfuric acid)
- **Longest rotation time** of all planets — one rotation (spin on its own axis) takes 243 Earth days
- One of only **two planets that rotate east to west** (the other is Uranus — see Page 4)

🧠 Fig 2: Mercury vs Venus – Key Facts



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💡 *Understand it like this...*

Venus is like a car parked with its heater stuck on full blast, windows sealed with acid-fog — CO_2 traps heat in, H_2SO_4 clouds keep it hazy — that's why it's both hottest and brightest.

📖 Rotation vs Revolution – quick refresher (tap to expand) ▼

🎯 *Exam Tip*

Venus has many titles — remember them as a set: **Lucifer, Earth's Twin, Morning/Evening Star, Hottest, Brightest, Longest rotation, East-to-west rotation.**

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Earth, Mars + the Asteroid Belt 3️⃣ Earth

- Known as the **Blue Planet**
- The densest planet in the solar system — density ≈ **$5.5\ g/cm^3$**
- The only planet where life is known to be possible
- Has one natural satellite — the **Moon** (we also send many artificial satellites via ISRO, NASA, and other space agencies, but the Moon is Earth's only natural

satellite)

4️⃣ Mars

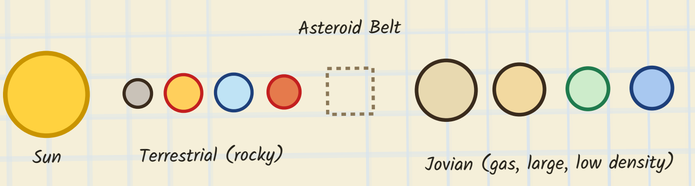
- Known as the **Red Planet** — red because of the presence of **iron oxide**
- Has **two satellites: Phobos** and **Deimos**
- Home to Olympus Mons — the **highest point found anywhere in the entire solar system**

> ❌ Common Mistake
> Earth has just **one** natural satellite (the Moon), while Mars has **two** (Phobos & Deimos) — don't swap these numbers!

Terrestrial vs Jovian Planets

The first four planets — Mercury, Venus, Earth, Mars — are called **terrestrial** or **inner planets**: they're rocky, with **high density** but **small size**. Beyond them lies the **Asteroid Belt** — a belt of very large rocks that separates the inner four planets from the outer four. The outer four — Jupiter, Saturn, Uranus, Neptune — are called **Jovian** or **outer planets**: made of gas, with **low density** but **large size**.

🧠 Fig 3: Layout of the Solar System



Small Bodies of the Solar System

Term	Meaning
Asteroid	Large rocky body, mainly found in the Asteroid Belt
Meteoroid	Small stone/rock piece travelling in space

Meteor	A meteoroid that enters Earth's atmosphere and burns up — appears as a "shooting star"
Meteorite	A meteoroid that survives and strikes Earth's surface
Comet	An icy object — as it passes near the Sun, its ice melts and it appears to be "burning"

🧠 *Memory Trick*

Meteoroid (in space) → **Meteor** (burns in our sky, "shooting star") → **Meteorite** (actually hits the ground).

Dwarf Planets

Ceres is a dwarf planet found within the Asteroid Belt.

🎯 *Exam Tip*

Density is **high** for terrestrial (rocky) planets but they're **small** in size; Jovian (gas) planets have **low** density but are **large** in size — an inverse relationship worth remembering.

Jovian Planets — Jupiter, Saturn, Uranus + Neptune

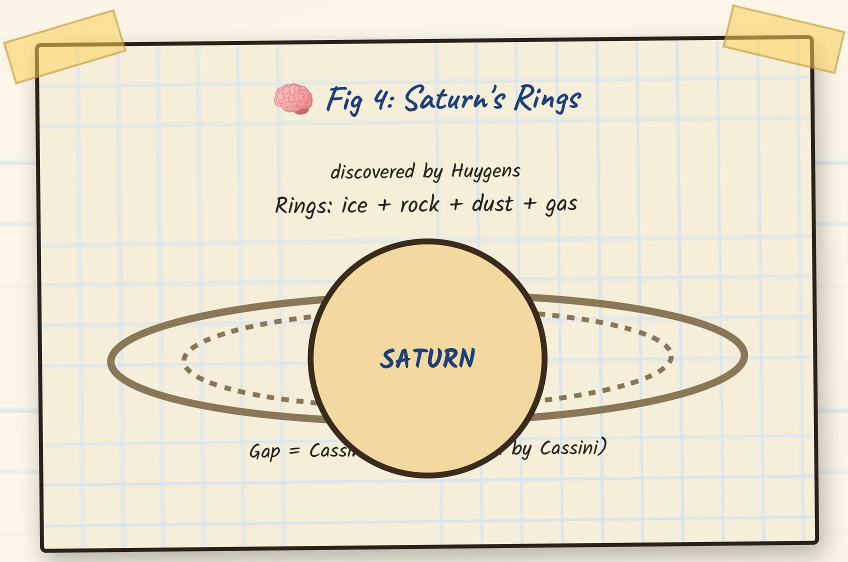
5️⃣ Jupiter

- The largest planet in the solar system
- **Shortest rotation time** of all planets — one rotation in just 9 hours 55 minutes
- Currently has the **most satellites** of any planet — though this exact count keeps changing as new moons are discovered, so this comparison has been dropped from exams

- *Largest moon:* ***Ganymede*** *— which is also the* ***largest satellite in the entire solar system***
- *The four* ***Galilean Moons*** *(discovered by* ***Galileo Galilei****):* ***Io, Europa, Ganymede,*** *and* ***Callisto***

6️⃣ Saturn

- *Famous for its* ***rings****, made of ice, rock, dust, and gases — rings exist on other outer planets too, but are most clearly visible on Saturn*
- ***Lowest density*** *of any planet — just* ***0.69 g/cm³****, so light that (hypothetically) Saturn would float if placed in water*
- *Largest satellite:* ***Titan***
- *Its rings were discovered by* ***Huygens****; the gap visible within the rings (the Cassini Division) was discovered by* ***Cassini***



🧠 *Fig 4: Saturn's Rings*

> ❌ **Common Mistake**
> Saturn is not the *only* ringed planet — all four Jovian planets have rings — Saturn's are just the most clearly visible.

7️⃣ Uranus

- *Known as the* ***Green Planet****, the* ***Rolling Planet****, and the* ***Lobsided Planet***
- *Its axis is tilted almost 98° — more than 90° — which makes it appear to "roll" as it moves*
- *Discovered by* ***William Herschel*** *in 1781*

- Holds the record for the **coldest temperature ever recorded** on any planet

8️⃣ Neptune

- The **farthest planet** from the Sun
- **Longest revolution time** of all planets — one revolution around the Sun takes 165 years

🧠 Memory Trick
"Green, Rolling, Lobsided" (Uranus, 98° tilt) → Herschel, 1781. "Farthest, Longest revolution" (Neptune, 165 yrs) — pair each planet with its one standout superlative!

Order, Dwarf Planets, Kuiper Belt + The Moon

Quick Comparison – Rotation & Revolution

Record	Planet	Value
Shortest rotation time	**Jupiter**	9 h 55 min
Longest rotation time	**Venus**	243 Earth days
Shortest revolution time	**Mercury**	88 Earth days
Longest revolution time	**Neptune**	165 years

❌ Common Mistake
Rotation (spin on its own axis) and **revolution** (orbit around the Sun) are different motions — don't mix up which record belongs to which!

Retrograde (East to West) Rotation

All planets in the solar system rotate **west to east** (counter-clockwise) — except **Venus and Uranus**, which are the only two planets that rotate **east to west** (clockwise).

Order of Planets by Size (Largest → Smallest)

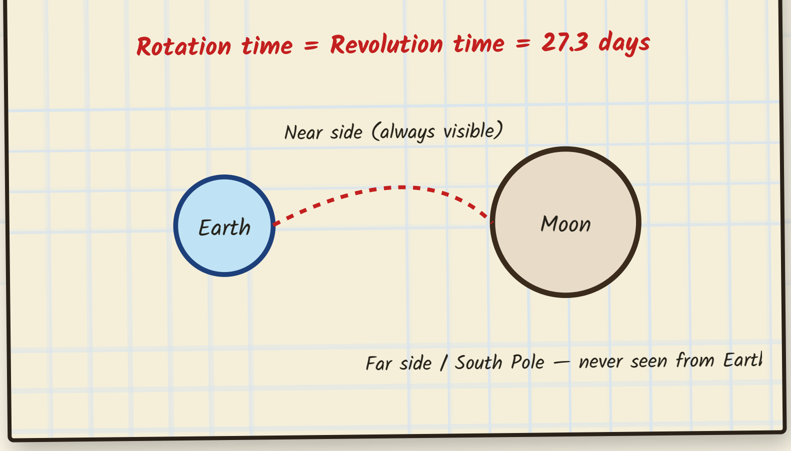
Jupiter → **Saturn** → **Uranus** → **Neptune** → **Earth** → **Venus** → **Mars** → **Mercury**

The four Jovian (outer) planets are all larger than any terrestrial (inner) planet. Also note: **Venus** is considered the planet **closest to Earth**.

Kuiper Belt & Dwarf Planets

Beyond Neptune lies the **Kuiper Belt**. Just as **Ceres** is a dwarf planet found in the Asteroid Belt, the Kuiper Belt is home to dwarf planets such as **Pluto, Haumea,** and **Makemake**.

Fig 5: The Moon's Synchronous Rotation



The Moon

- **Non-luminous** — reflects the Sun's light
- Only **one side of the Moon** is ever visible from Earth, because its **rotation time and revolution time are equal** — both are 27.3 days
- The side that's never visible is called the **far side** (or the South Pole region) of the Moon

- **Chandrayaan-3:** India became the **4th country** to achieve a soft landing near the Moon's South Pole
- The Moon's gravity is **1/6th of Earth's gravity** — mass doesn't change, only **weight** changes (since weight depends on gravity, not mass)
- Highest point on the Moon: **Mons Huygens**

🚀 Chandrayaan-3 made India the 4th nation ever to soft-land near the Moon's South Pole — a region we can never see from Earth!

🎯 Exam Tip

Same number, two meanings: the Moon's rotation time = revolution time = **27.3 days** — that's exactly why we only ever see one face of the Moon.

⭐ Exam-Ready Summary

- **Closest to Sun:** Mercury | **Farthest from Sun:** Neptune
- **Hottest & brightest:** Venus | **Densest:** Earth (5.5 g/cm³) | **Lowest density:** Saturn (0.69 g/cm³)
- **Largest planet:** Jupiter | **Largest moon in solar system:** Ganymede
- **Rotate East→West:** only Venus & Uranus
- **Asteroid Belt** separates terrestrial planets from Jovian planets; **Kuiper Belt** lies beyond Neptune